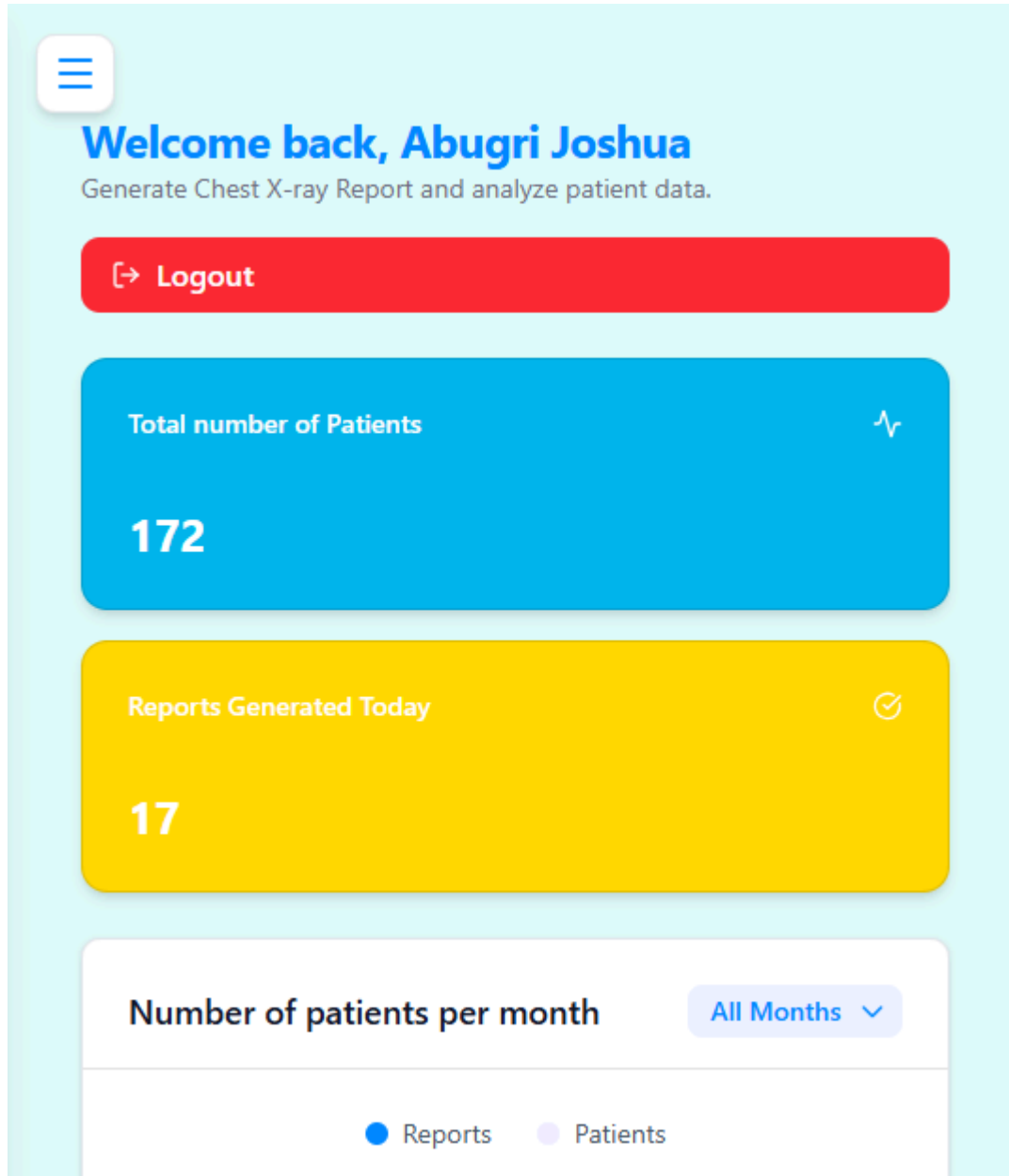
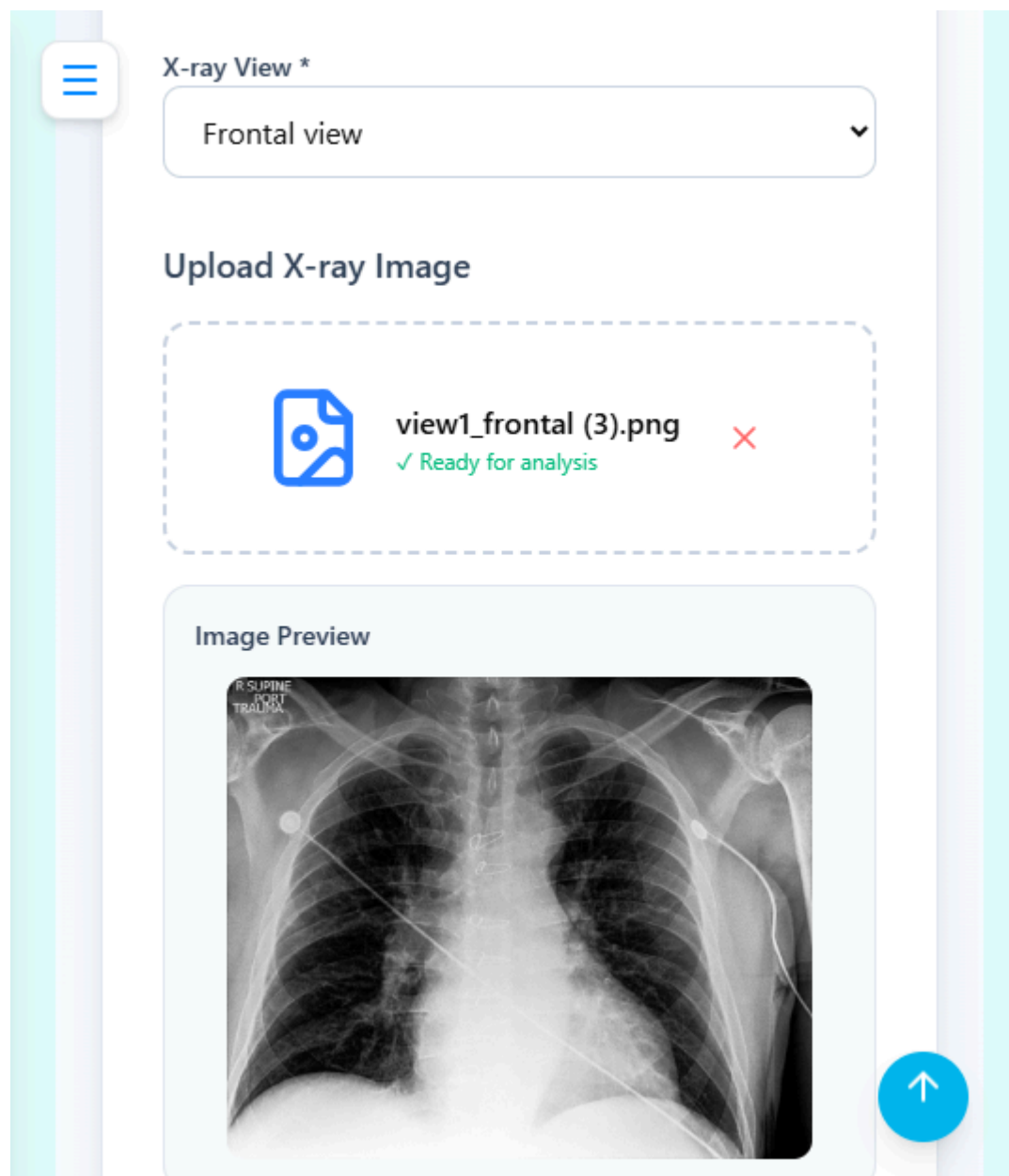


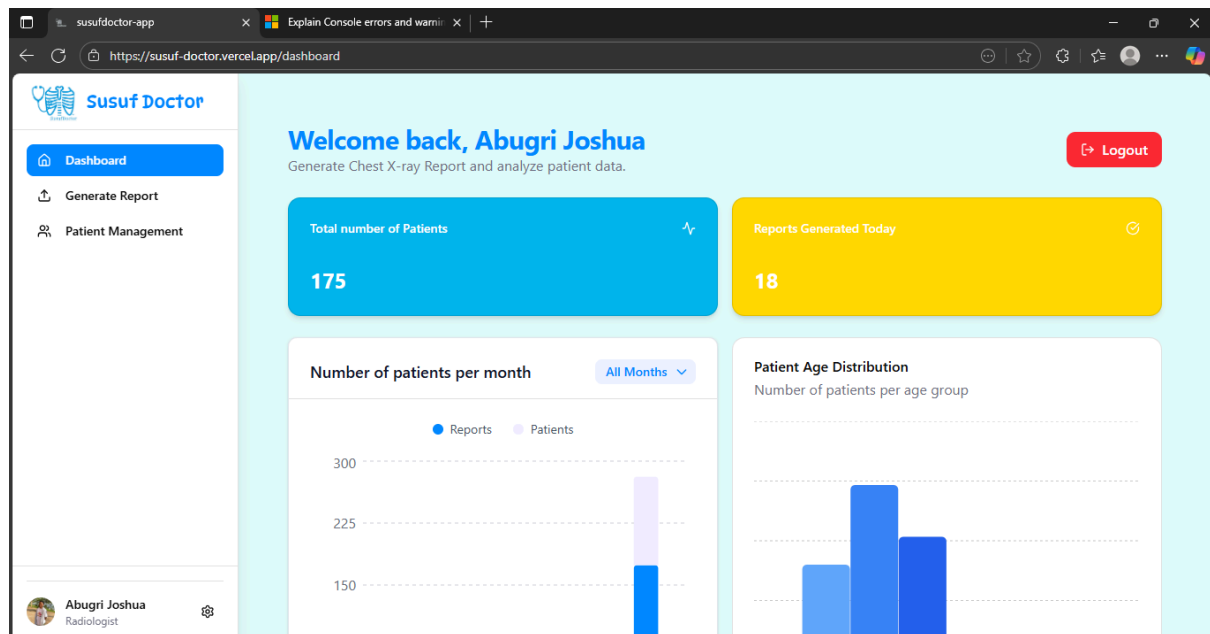
Final version of the product/solution

1. Testing Across different screen sizes





The application was also tested on the Edge browser and it works perfectly



Unit test was also conducted to test the application

Dashboard vitest: this was to ensure that all the functionalities of the dashboard are fully functional.

```
> assets
> components
> dashboard
> utils
App.jsx
# index.css
main.jsx
setupTests.js
test
  dashboard.test.jsx
  signup.test.jsx
.gitignore
eslint.config.js
index.html
package-lock.json
package.json
postcss.config.js
README.md
tailwind.config.js
vercel.json
vite.config.js
```

```
HP@AKTNPANG MINGM64 ~/Desktop/SusufDoctor/SusufDoctor/susufdoctor-app (main)
$ npm test dashboard.test.jsx
> susufdoctor-app@0.0.0 test
> vitest dashboard.test.jsx

DEV v4.0.9 C:/Users/HP/Desktop/SusufDoctor/SusufDoctor/susufdoctor-app

✓ test/dashboard.test.jsx (12 tests) 500ms
  ✓ DashboardPage Component (12)
    ✓ renders dashboard page by default 59ms
    ✓ loads current page from localStorage on mount 10ms
    ✓ saves current page to localStorage when page changes 17ms
    ✓ navigates between pages using navigation buttons 16ms
    ✓ navigates when Dashboard component calls onNavigate 6ms
    ✓ shows scroll-to-top button when scrolling down 212ms
    ✓ hides scroll-to-top button when scrolling to top 70ms
    ✓ scrolls to top when scroll-to-top button is clicked 36ms
    ✓ cleans up scroll event listener on unmount 37ms
    ✓ renders all page options in navigation 3ms
    ✓ handles invalid page names by showing dashboard 2ms
    ✓ maintains scroll position state separately from page state 27ms

Test Files 1 passed (1)
Tests 12 passed (12)
Start at 15:03:51
Duration 3.90s (transform 430ms, setup 277ms, collect 931ms, tests 500ms, environment 1.70s, prepare 35ms)
```

Sign up: This testing was to ensure authentication is working properly

```
public
├── src
│   ├── assets
│   ├── components
│   ├── dashboard
│   ├── utils
│   ├── App.jsx
│   ├── index.css
│   ├── main.jsx
│   ├── setupTests.js
│   └── test
│       ├── dashboard.test.jsx
│       └── signup.test.jsx
├── .gitignore
├── eslint.config.js
├── index.html
├── package-lock.json
├── package.json
├── postcss.config.js
├── README.md
├── tailwind.config.js
├── vercel.json
└── vite.config.js

TERMINAL
HP@WINPANG MINGW64 ~/Desktop/SusufDoctor/SusufDoctor/susufdoctor-app (main)
$ npm test signup.test.jsx
> susufdoctor-app@0.0.0 test
> vitest signup.test.jsx

DEV v4.0.9 C:/Users/HP/Desktop/SusufDoctor/SusufDoctor/susufdoctor-app

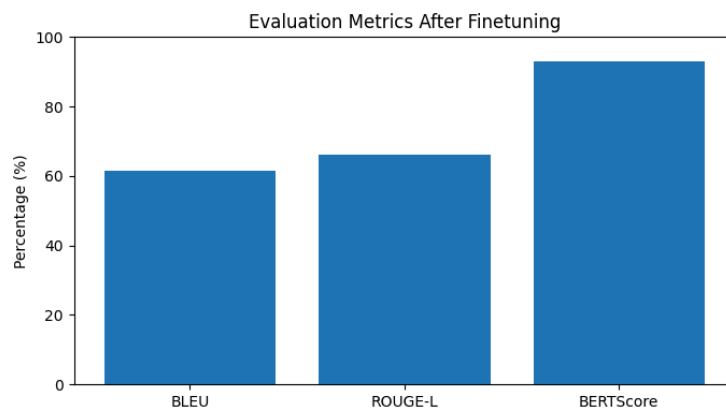
stderr | test/signup.test.jsx > Signup Component > renders signup form fields
▲React Router Future Flag Warning: React Router will begin wrapping state updates in `React.startTransition` in v7. You can use the `startTransition` future flag to opt-in early. For more information, see https://reactrouter.com/v6/upgrading/future#v7_starttransition.
▲React Router Future Flag Warning: Relative route resolution within Splat routes is changing in v7. You can use the `v7_relativeSplatPath` future flag to opt-in early. For more information, see https://reactrouter.com/v6/upgrading/future#v7_relativeSplatPath.

✓ test/signup.test.jsx (6 tests) 562ms
  ✓ Signup Component (6)
    ✓ renders signup form fields 76ms
    ✓ shows validation errors when submitting empty form 41ms
    ✓ toggles password visibility 72ms
    ✓ switches from signup to login mode 345ms
    ✓ submits signup data successfully 32ms
    ✓ shows API error message on failed signup 43ms

Test Files 1 passed (1)
Tests 6 passed (6)
Start at 15:02:08
Duration 6.08s (transform 256ms, setup 334ms, collect 675ms, tests 562ms, environment 3.18s, prepare 28ms)
```

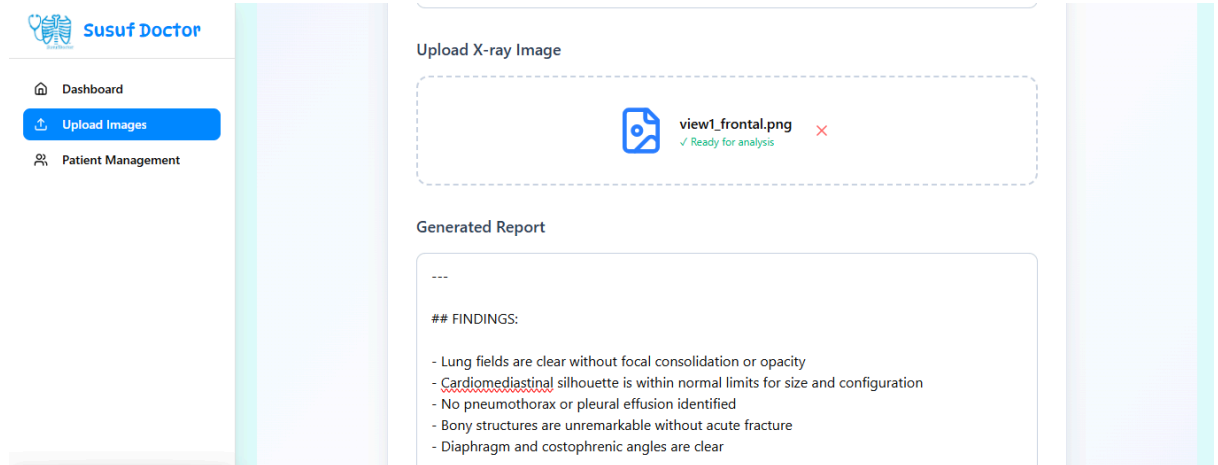
2. Analysis of the results

The SuSufDoctor Application was evaluated to determine how well it met the project objectives and how effectively it performed in generating accurate, longitudinal chest X-ray reports. The fine-tuned SmolVLM-500M model achieved significant improvements across all evaluation metrics. BLEU increased from 1.29% to 61.53%, ROUGE-L from 8.26% to 66.08%, and BERTScore (F1) from 80.48% to 93.92%. Training stabilized by the third epoch, confirming rapid convergence. The deployed system generated structured radiology reports within 5–30 seconds and was successfully tested by five radiologists, who expressed strong satisfaction with its accuracy and usability.

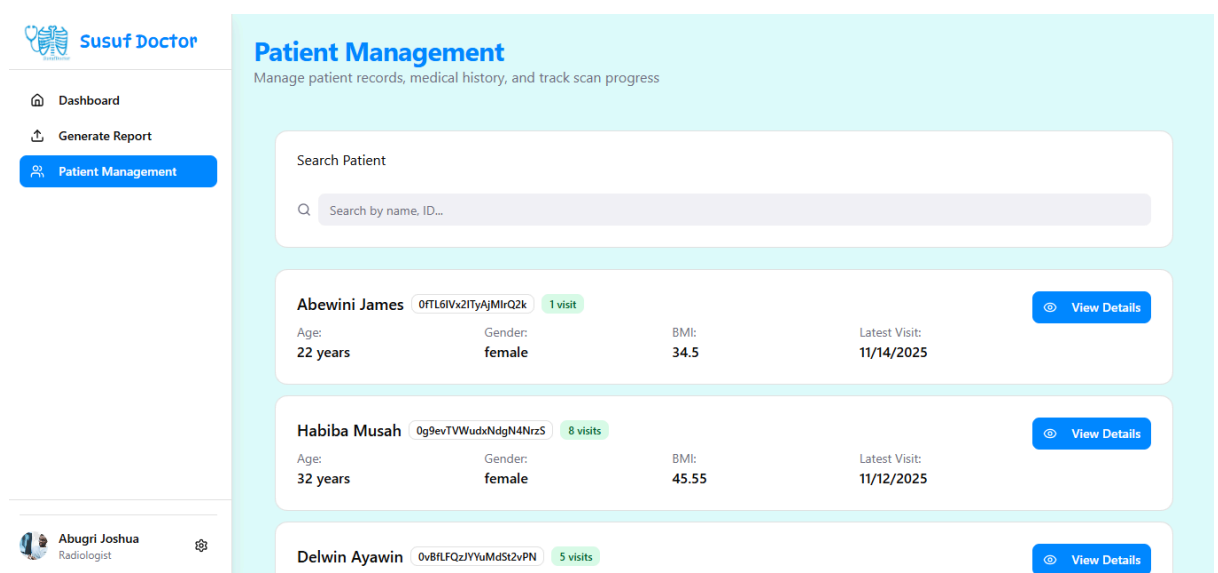


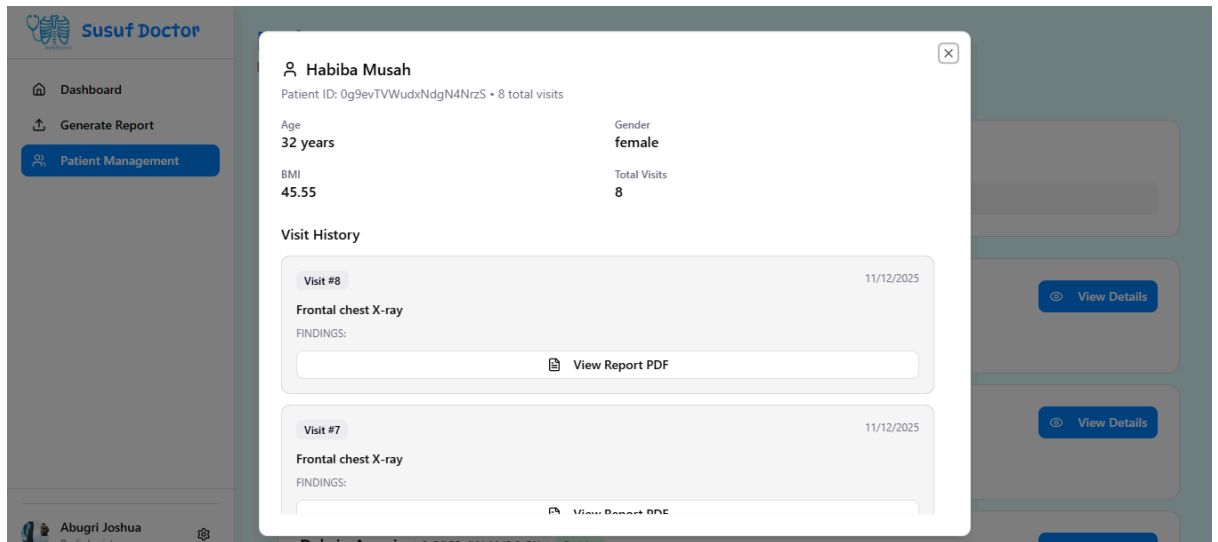
These results were obtained through longitudinal dataset preparation, where over 1,500 paired studies were cleaned, aligned, and formatted. LoRA fine-tuning allowed the model to learn radiology-specific patterns efficiently, while 4-bit quantization enabled training on limited hardware. The multimodal architecture supported fusion of current images, prior images, and patient metadata, improving temporal reasoning.

All project objectives were achieved: preparation of a longitudinal dataset, development of a multimodal vision language model, generation of complete radiology reports, deployment in a functional web application, and validation by real users. The system performs reliably across these tasks and produces structured, clinically meaningful outputs.

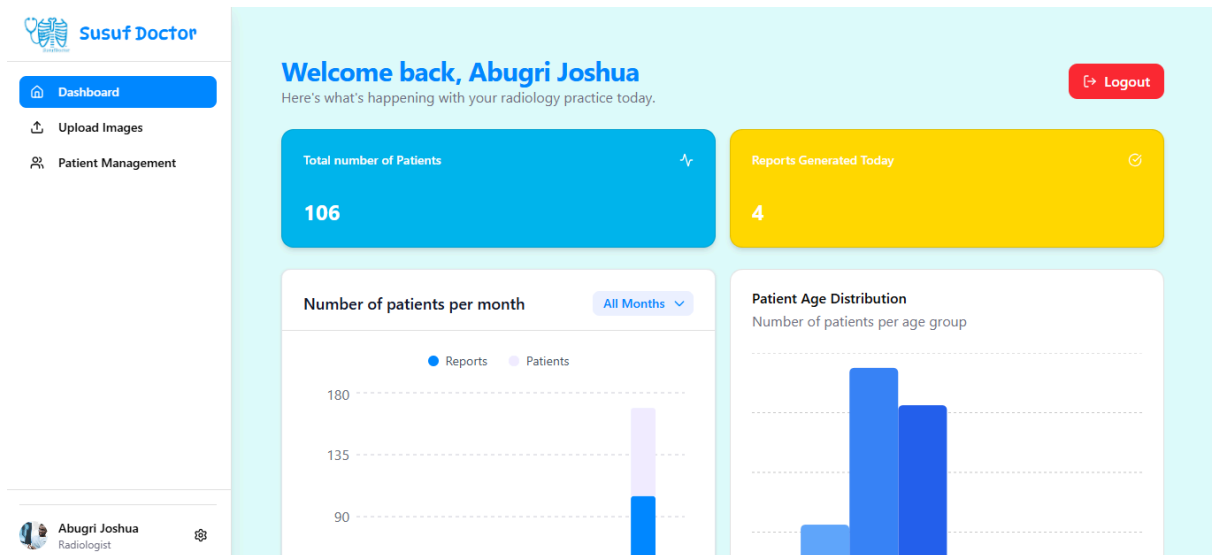


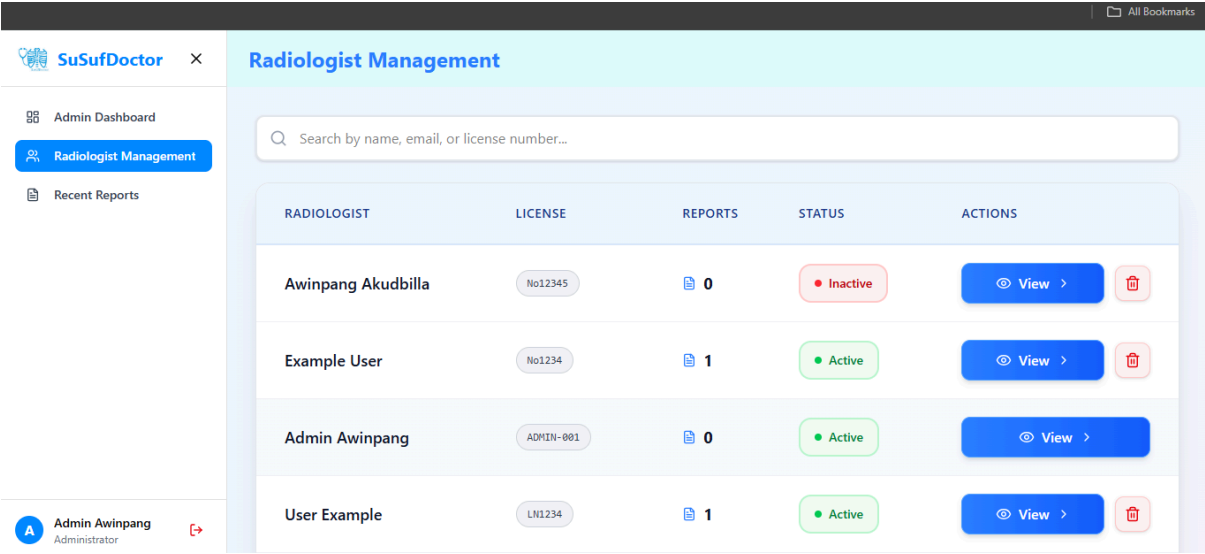
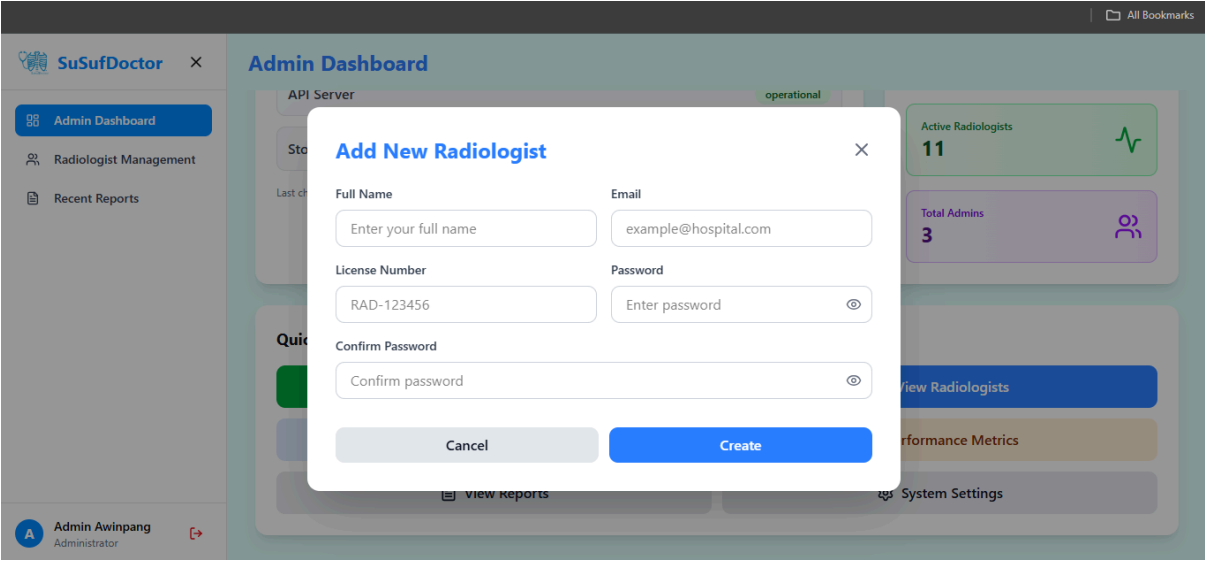
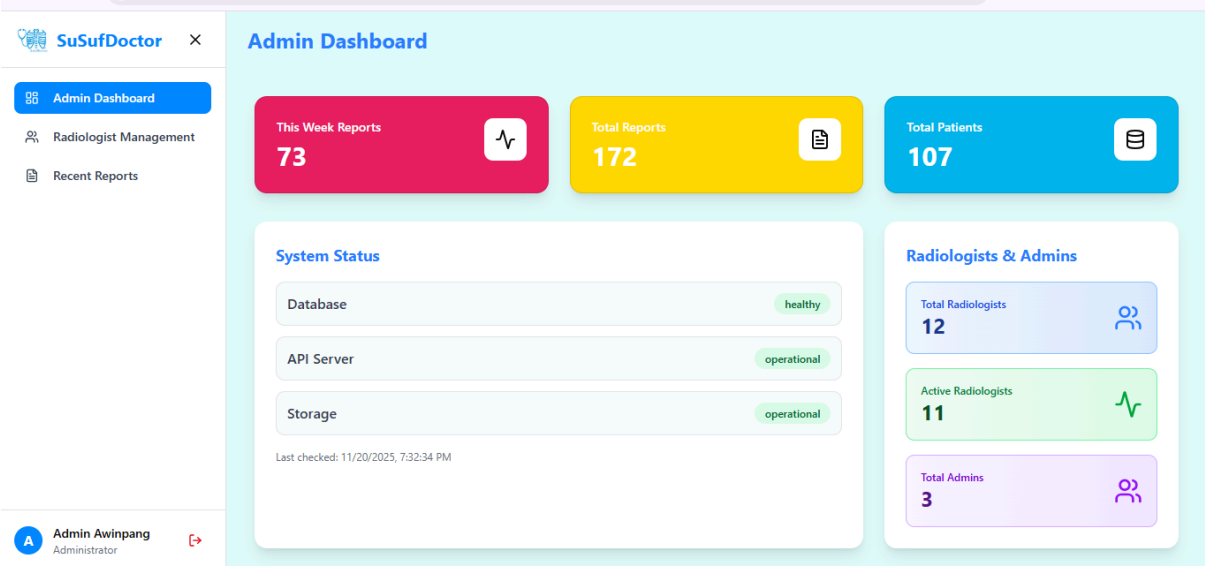
In addition to the model's performance, the SuSufDoctor application offers several important features that support radiologists in practice. The report editing feature allows users to refine generated text by adding or removing clinical details, ensuring that the final output aligns with professional judgment; this capability was highly valued by the radiologists who tested the system. The patient management module provides access to patient records, previous visits, and past reports, enabling quick retrieval of longitudinal histories.





The analytics dashboard summarizes key statistics such as patient distribution, report frequency, and age or gender breakdowns, giving users an overview of imaging trends. The word cloud feature highlights the most frequently occurring terms across generated reports, offering a simple way to visualize common diagnostic patterns. Together, these features align directly with the project objectives and demonstrate that the system not only generates reports but also supports clinical workflow, usability, and decision-making.





SuSufDoctor

Admin Dashboard

Radiologist Management

Recent Reports

Admin Awinpang

Administrator

Radiologist Management

Abugri Joshua

Radiologist Profile

EMAIL ADDRESS

joshua@gmail.com

ROLE

User

STATUS

Active

LICENSE

No1234

REPORTS

64

Deactivate

Make Admin

Delete Radiologist

Patients

22 Total

Abewini James

9 visits

ID: 0fTL6IVx2ITyAjMIRQ2k

Age: 22

Sex: Female

Awinbila

2 visits

ID: Bz8ePOFqWe9Z1K9oo3Bq

Age: 30

Sex: Female

Awinbila Ayawin

1 visits

ID: 1y4A1z6VPHkcxPmD1ZFf

Age: 30

Sex: Female

SuSufDoctor

Admin Dashboard

Radiologist Management

Recent Reports

Admin Awinpang

Administrator

Radiologist Management

Awinbila

Patient Profile

PATIENT ID

Bz8ePOFqWe9Z1K9oo3Bq

AGE

30 years

SEX

Female

BMI

45.567

2

Total Visits

Visit History

2 Visits

Frontal chest X-ray

Visit #1

November 13, 2025

impression: single frontal chest radiograph demonstrates stable position of the left subclavian venous catheter and a right internal jugular central venous catheter with the tip in the mid supine spine. no pneumothorax or edema is identified. there are bilaterally mild reticulation consistent with atelectasis.

Download Report

View X-ray

Frontal chest X-ray

Visit #2

November 13, 2025

HISTORY: Female with shortness of breath. COMPARISON: Chest x-ray dated 12/28/IMPRESSON: SINGLE PORTABLE CHEST RADIOGRAPH DEMONSTRATES STABLE POSITION OF THE LEFT SUBCLAVIAN VENOUS CATHETER AND A RIGHT IJ CENTRAL VENOUS CATHETER WITH THE TIP IN THE MID SUPINE SPINE. NO PNEUMOTHORAX OR EDEMA IS IDENTIFIED. THERE ARE BILATERALLY MILD RETICULATION CONSISTENT WITH ATELECTASIS.

 SuSufDoctor

Admin Dashboard

Radiologist Management

Recent Reports

Admin Awinpang

Administrator

Radiologist Management

User Example	No2345	4	Active	View	
Ateiwini Awiniini		0	Active	View	
Abugri Joshua		0	Active	View	
User3 Example		0	Active	View	
Awinpiini Adelwin	LIC123456	0	Active	View	
Abugri Ayawin	No1234	0	Active	View	

Confirm Deletion

Are you sure you want to delete Ateiwini Awiniini? This action cannot be undone.

CancelDelete

 SuSufDoctor

Admin Dashboard

Radiologist Management

Recent Reports

Admin Awinpang

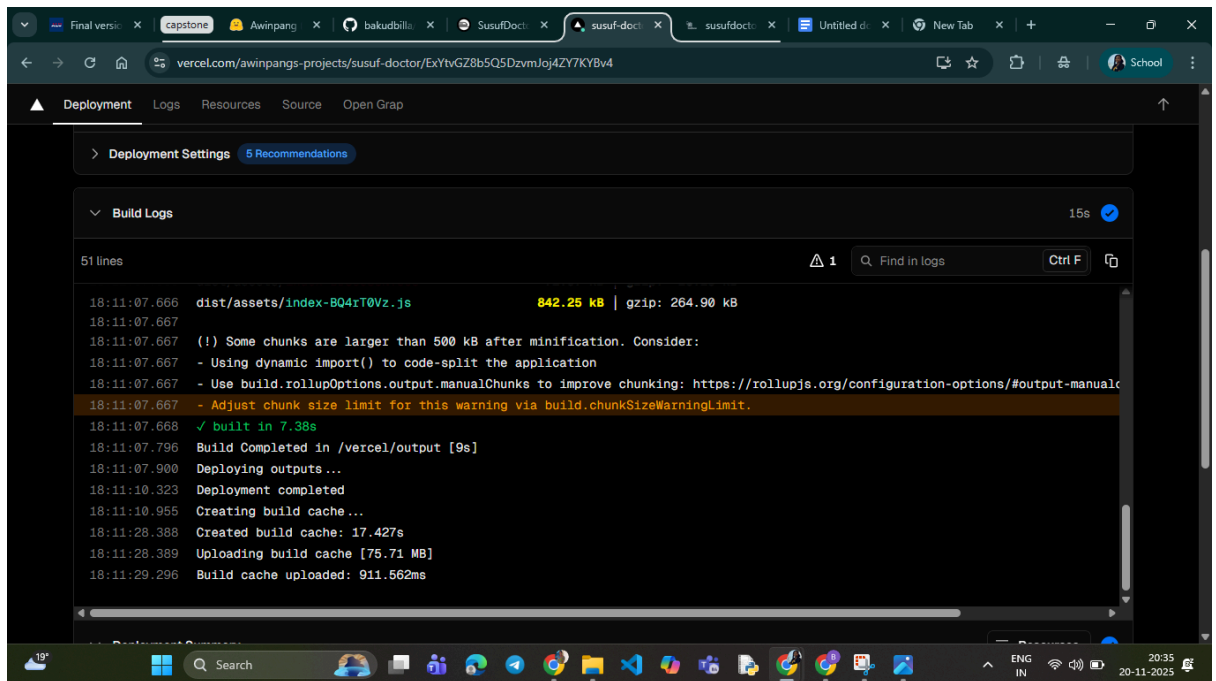
Administrator

Recent Reports

Patient	Radiologist	View Type	Date	Status	Action
Joy Akurugu	Abugri Joshua	Frontal	11/20/2025	Generated	View
Delwin Ayawin	Abugri Joshua	Frontal	11/20/2025	Generated	View
Akora Ama	Abugri Joshua	Frontal	11/20/2025	Generated	View
John Aj	Abugri Joshua	Frontal	11/20/2025	Generated	View
Evans Atta	Abugri Joshua	Other	11/20/2025	Generated	View
John Aj	Abugri Joshua	Other	11/20/2025	Generated	View
Jay	Abugri Joshua	Frontal	11/20/2025	Generated	View

3. System Deployment

Frontend Deployment on vercel: <https://susuf-doctor.vercel.app/>



FAST API Backend Deployment:

<https://susufdoctorbackend-100056520598.europe-west1.run.app/docs>

Dashboard

History

Repositories

Triggers

Worker pools

Permissions

Release notes

Build details

Retry build

Copy URL

Learn

Successful: Od4efa5a-826a-4ea8-a75b-633c05c8216f

Started on 20 Nov 2025, 18:03:09

Trigger

rmgpgab-susufdoctorbackend-europe-west1-bakudbilla-SusufDoctwlo

Source

bakudbilla/SusufDoctor

Branch

main

Commit

1a2c001

Steps	Duration
Build summ... 3 Steps	00:20:17
0: Buildpack pack build eur...	00:10:15
1: Push push europe-w...	00:06:35
2: Deploy gcloud run ser...	00:03:16

Build summary

Build log

Execution details

Build artifacts

Controls

Expand

View in Logs Explorer

1

starting build "0d4efa5a-826a-4ea8-a75b-633c05c8216f"

2

3

FETCHSOURCE

4

From https://github.com/bakudbilla/SusufDoctor

5

* branch 1a2c001f1c11ce9b290f796bebea453d59e214 -> FET

6

HEAD is now at 1a2c001 patient form

7

GitCommit:

8

1a2c001f1c11ce9b290f796bebea453d59e214

9

BUILD

A clear and well-structured deployment plan was developed to ensure that the SuSufDoctor system could operate reliably in a production environment. The deployment process documented all required tools, environments, and configurations, covering both the backend and frontend components. The system was successfully deployed in the intended environment, and functionality was verified through comprehensive testing. End-to-end evaluations including image upload, report generation, patient management, report editing, and analytics confirmed that the deployed system performs correctly and consistently in the target environment.

Step by Step Deployment Plan

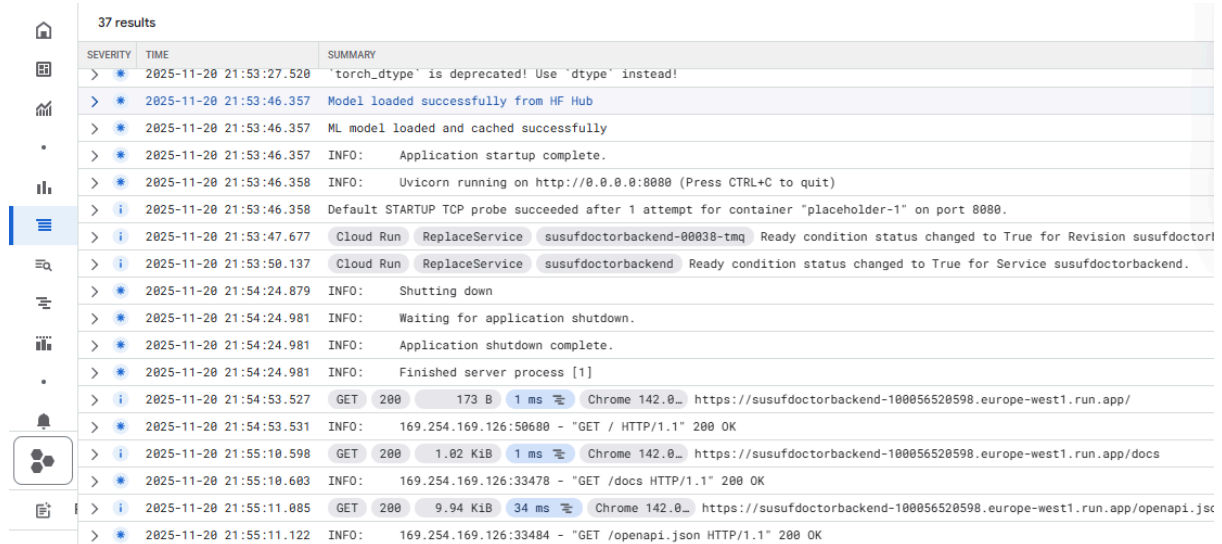
Frontend Deployment

The frontend of the SuSufDoctor system, located in the `susufdoctor-app` directory on the github, was deployed using the vercel hosting platform. Before deployment, all environment variables such as API endpoints and Firebase configuration keys were added to the project's `.env` file to ensure correct communication with the backend services. The React application was built using `npm run build`, generating a production-ready build folder optimized for performance. This build folder was uploaded to Vercel as a static site, where the platform automatically configured hosting settings, caching, and SSL support. Once deployed, the frontend was fully accessible through the generated vercel URL, and all core features such as image upload, report generation, patient management, analytics, word-cloud visualization, and editing were successfully tested in the live environment. The deployment on Vercel ensured reliable availability, automatic updates on new commits, and consistent integration with the backend API.

Backend Deployment

The backend of the SuSufDoctor system, developed with FastAPI, was deployed on Google Cloud Run using Buildpacks, which allowed the application to be containerized automatically without requiring a Dockerfile. The backend source code was connected directly to Google Cloud, and Cloud Build generated a production-ready container image using Buildpacks. This image was then deployed to Cloud Run as a fully managed, serverless service. All required environment variables such as Firebase credentials, bucket configurations, and API keys were securely added through the Cloud Run console. Cloud Run handled HTTPS, automatic scaling, concurrency, and traffic management without manual configuration. After

deployment, the backend URL was integrated with the frontend, and all major backend features including model inference, image upload handling, authentication, and patient management were successfully tested in the live environment, confirming that the system operated as intended.



The screenshot displays the Google Cloud Logs interface. On the left, there is a sidebar with navigation icons: Home, Logs Explorer, Alerts, Dashboards, and Settings. The main panel shows a list of log entries for the service 'susufdoctorbackend'. The table has columns for SEVERITY, TIME, and SUMMARY. The log entries include messages about model loading, application startup, and HTTP requests. The first entry is a warning about a deprecated 'torch_dtype' parameter. Subsequent entries show successful model loading and application startup. The final entries show HTTP GET requests to the application's root, /docs, and /openapi.json endpoints, all returning 200 OK status.

SEVERITY	TIME	SUMMARY
> *	2025-11-20 21:53:27.520	"torch_dtype" is deprecated! Use "dtype" instead!
> *	2025-11-20 21:53:46.357	Model loaded successfully from HF Hub
> *	2025-11-20 21:53:46.357	ML model loaded and cached successfully
> *	2025-11-20 21:53:46.357	INFO: Application startup complete.
> *	2025-11-20 21:53:46.358	INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
> i	2025-11-20 21:53:46.358	Default STARTUP TCP probe succeeded after 1 attempt for container "placeholder-1" on port 8080.
> i	2025-11-20 21:53:47.677	Cloud Run ReplaceService susufdoctorbackend-00038-tmq Ready condition status changed to True for Revision susufdoctorb...
> i	2025-11-20 21:53:50.137	Cloud Run ReplaceService susufdoctorbackend Ready condition status changed to True for Service susufdoctorbacken...
> *	2025-11-20 21:54:24.879	INFO: Shutting down
> *	2025-11-20 21:54:24.981	INFO: Waiting for application shutdown.
> *	2025-11-20 21:54:24.981	INFO: Application shutdown complete.
> *	2025-11-20 21:54:24.981	INFO: Finished server process [1]
> i	2025-11-20 21:54:53.527	GET 200 173 B 1 ms Chrome 142.0... https://susufdoctorbackend-100056520598.europe-west1.run.app/
> *	2025-11-20 21:54:53.531	INFO: 169.254.169.126:50680 - "GET / HTTP/1.1" 200 OK
> i	2025-11-20 21:55:10.598	GET 200 1.02 KiB 1 ms Chrome 142.0... https://susufdoctorbackend-100056520598.europe-west1.run.app/docs
> *	2025-11-20 21:55:10.603	INFO: 169.254.169.126:33478 - "GET /docs HTTP/1.1" 200 OK
> i	2025-11-20 21:55:11.085	GET 200 9.94 KiB 34 ms Chrome 142.0... https://susufdoctorbackend-100056520598.europe-west1.run.app/openapi.jsc
> *	2025-11-20 21:55:11.122	INFO: 169.254.169.126:33484 - "GET /openapi.json HTTP/1.1" 200 OK

Note: Please refer to [README.md](#) for setting up on local host
Demo [Link](#)